peaks; experiments confirm that supposition. But where is the particle? Since the wave is uniformly spread throughout space, there is no way for us to say the electron is *here* or *there.* When measured, it literally could be found anywhere. So, while we know precisely how fast the particle is moving, there is huge uncertainty about its position. And as you see, this conclusion does not depend on our disturbing the particle. We never touched it. Instead, it relies on a basic feature of waves: they can be spread out.

Although the details get more involved, similar reasoning applies to all other wave shapes, so the general lesson is clear. In quantum mechanics, uncertainty just is.

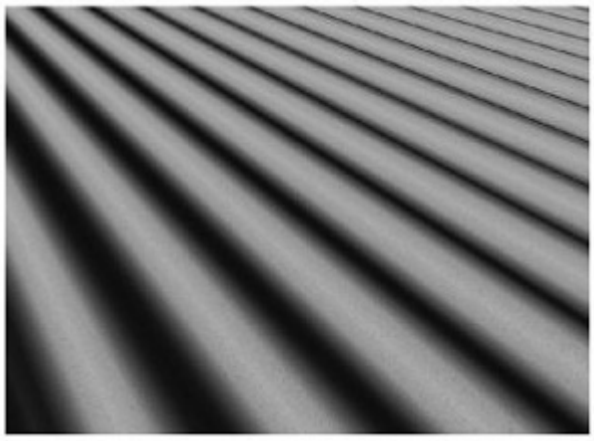


Figure 4.6 A probability wave with a uniform succession of peaks and troughs represents a particle with a definite velocity. But since the peaks and troughs are uniformly spread in space, the particle's position is completely undetermined. It has an equal likelihood of being anywhere.

Einstein, Uncertainty, and a Question of Reality

An important question, and one that may have occurred to you, is whether the uncertainty principle is a statement about what we can know about reality or whether it is a statement about reality itself. Do objects making up the universe really have a position and a velocity, like our usual classical image of just about everything—a soaring baseball, a jogger on the boardwalk, a sunflower slowly tracking the sun's flight across the sky—although quantum uncertainty tells us these features of